Chronological Operation of the Two-Board LDA Classifier

Simulated waveforms, 2 boards × 4 channels, real FORS-EMG (subject 6, trial 1): from reset to a decision

Reset + identity
0 – 40 µs
steps 1 – 2

→ Configuration
171 writes, 40 µs – 1.38 ms
steps 3

→ core_en → run
1.38 ms
steps 4 – 5

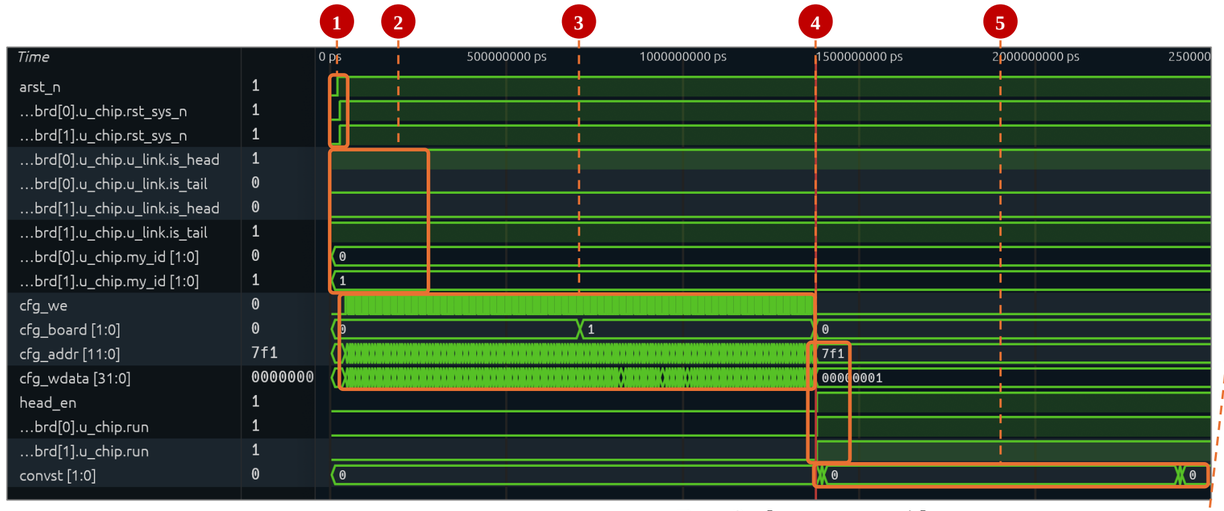
→ Sampling
one frame per 1.016 ms
steps 7

→ Stride ends
every 98 frames (99.5 ms)
steps 6, 8

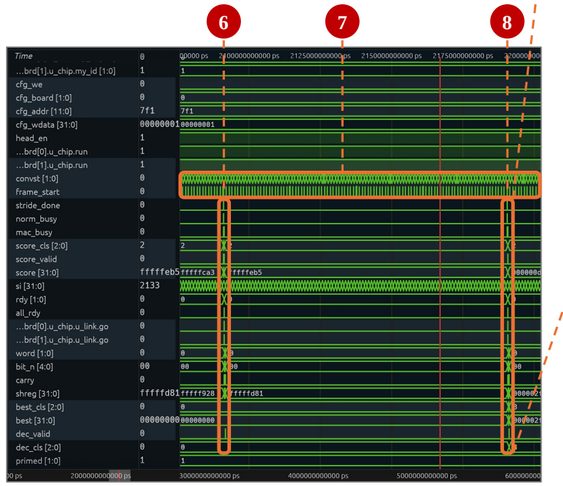
→ Stage 2 → Stage 3 → chain
→ decision (≈ 0.75 ms)
steps 9 – 15

repeats every stride

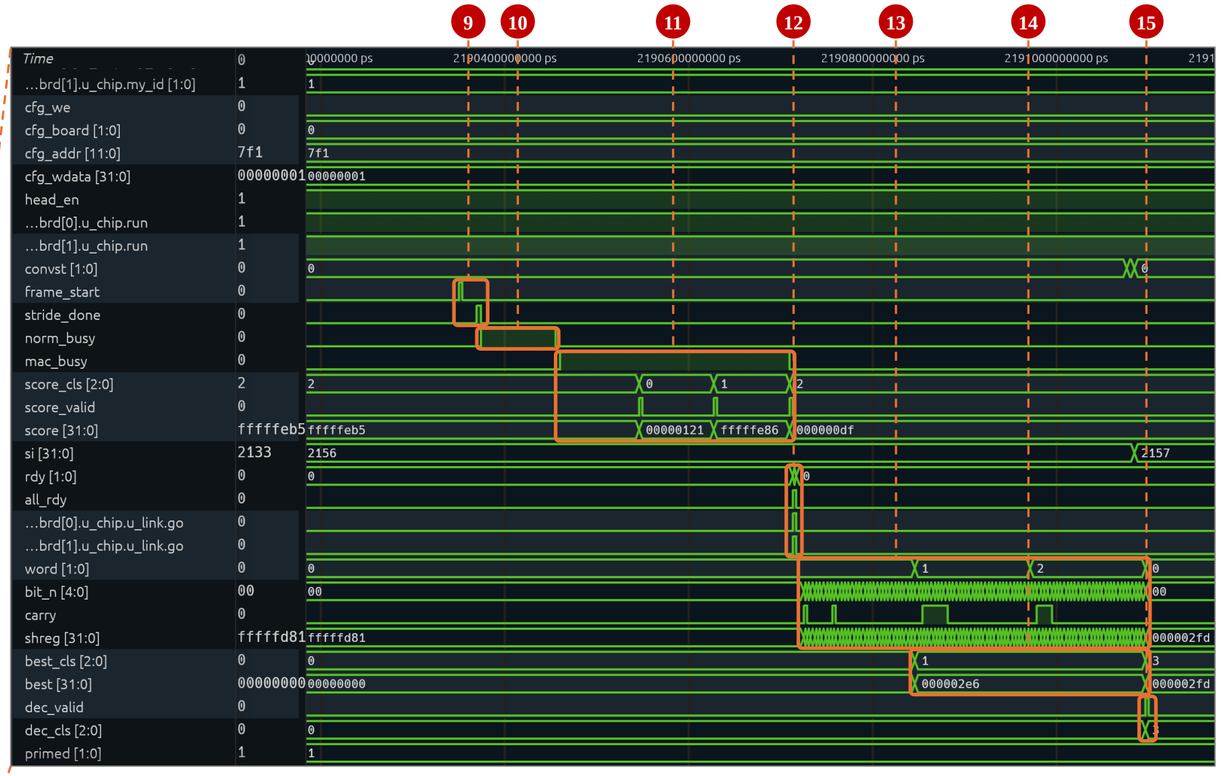
A Start-up (0 – 2.5 ms)

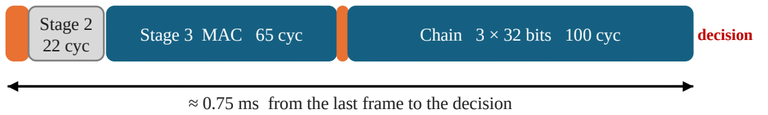


B ≈ 2 s later: two strides



C Zoom on one stride, from its last frame (≈ 1 ms)





Steps

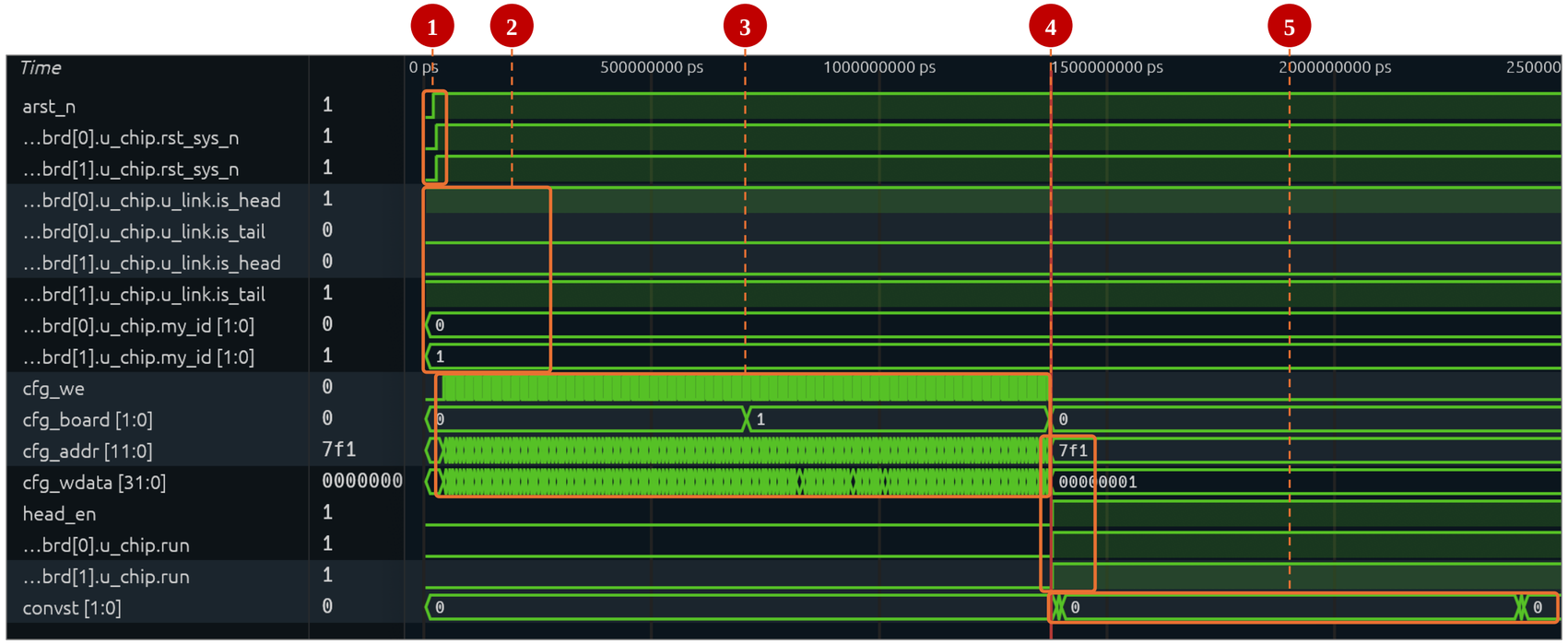
1. Reset released (20 µs)
2. Head / tail and ids found from pins
3. 171 config writes, board 0 then board 1
4. core_en → run on both boards (1.38 ms)
5. CONVST every 1.016 ms, in lockstep
6. Stride ends: one burst of work
7. 98 frames of sampling (99.5 ms)
8. Next stride: dec_cls 0 → 3
9. Last frame in → stride_done
10. Stage 2 normalise, 22 cycles
11. Stage 3 MAC, 65 cycles, 3 partials
12. rdy → all_rdy → go (barrier)
13. Serial add along the chain, 96 bits
14. Running argmax: D1 742, D2 −255, D3 765
15. Decision: class 3

Bit-exact with the golden model on all 79 decisions, for six board splits (verify_all.py: 18/18 checks pass).

Stride 21 (samples 1960–2155): D = head + tail
D1 = 289 + 453 = 742
D2 = −378 + 123 = −255
D3 = 223 + 542 = 765
argmax{0, D1, D2, D3} = class 3

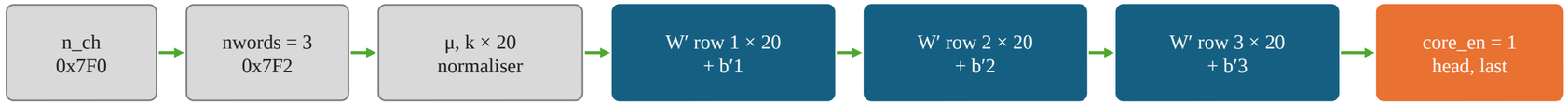
A · Start-up: reset → identity → configuration → run

No board is told where it is: position and id come from two pins; the host only writes the model





What each board receives (in this order)



4 + 2 + 20 + 3 × 21 = 85 writes per board; 2 × 85 + 1 = 171 writes, 2 clocks each ≈ 1.34 ms.
b′ goes to the head only (the tail gets 0), so the sum of the partials equals the single-chip LDA.

1 Reset
arst_n rises at 20 µs; rst_sys_n follows on both boards.

2 Self-identification
From two pins only: board 0 finds is_head, board 1 is_tail. The id ripples down the chain: my_id = 0, 1.

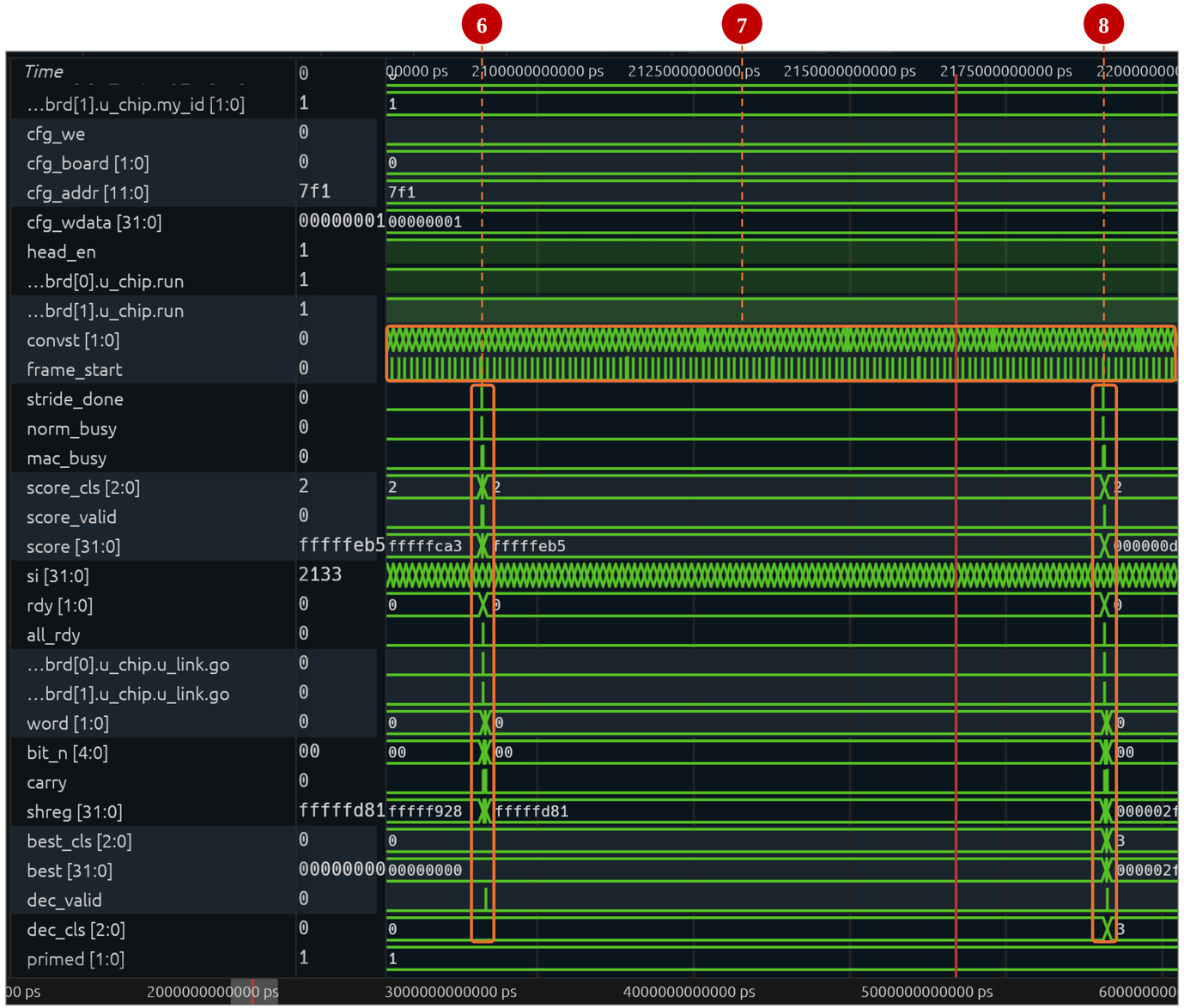
3 Configuration
The host loads the model on the shared bus: 171 writes in ≈ 1.34 ms. cfg_board selects the board by id: 85 writes to board 0, then 85 to board 1.

4 Start
Last write 0x7F1 = 1 is the head's core_en. head_en rises at 1.38 ms and run reaches both boards in the same cycle.

5 Lockstep sampling
Both CONVST timers start together: one 8-channel frame every 260 clocks = 1.016 ms (984.6 Hz).

B · Steady state: one burst of work per stride

The board only samples for 99.5 ms, then classifies in under 1 ms and goes back to sampling



6 **Stride ends**

After every 98th frame one short burst does all the work: stride_done → Stage 2 → Stage 3 → rdy / go → chain → dec_valid. Here decision 19 at 2.0916 s: class 0.

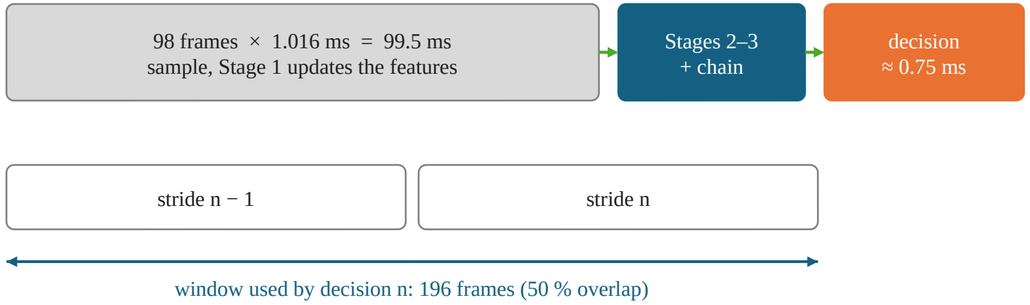
7 **Sampling**

For the next 98 frames (99.5 ms) only CONVST, the frame read and Stage 1 run; si counts the samples and dec_cls holds the last class.

8 **Next stride**

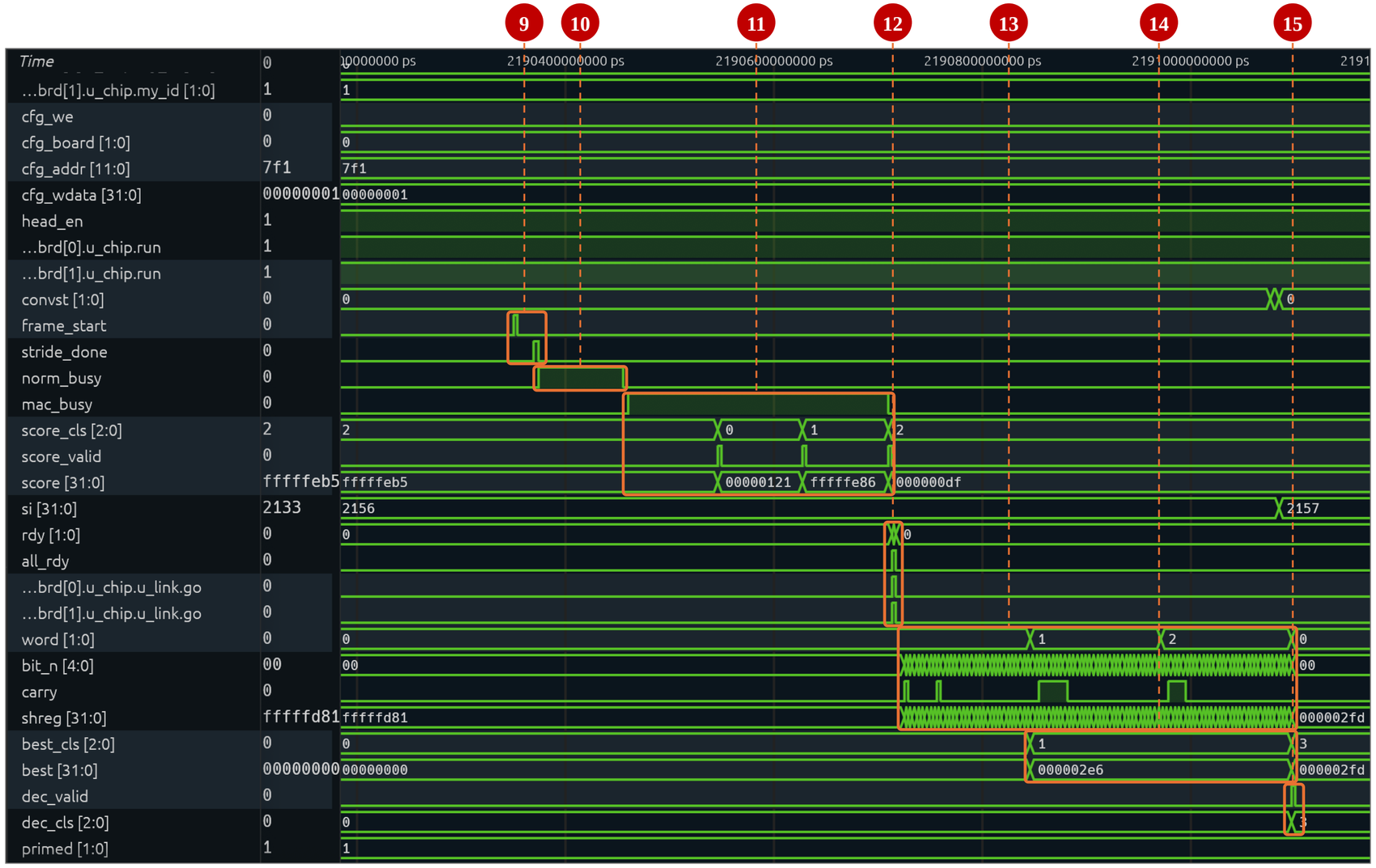
Decision 20 at 2.1911 s: dec_cls changes 0 → 3. primed = 1, so every decision after the first is valid. This burst is opened up in C.

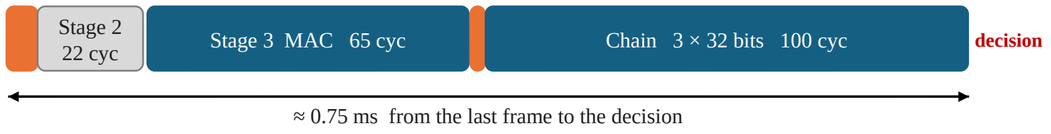
One stride, not to scale



C · Inside one stride: from the last frame to the decision

All boards compute in parallel on their own channels; only three partial sums travel along the chain





9 **Last frame of the stride**
frame_start: the 98th frame is in; Stage 1 updates the window features and raises stride_done.

10 **Stage 2: normalise**
norm_busy: the 20 features of this board, 22 cycles (86 µs).

11 **Stage 3: W′ · q**
mac_busy: 3 rows × 21 cycles (254 µs). One partial per row on score_valid: 289, −378, 223 (head, with b′).

12 **Barrier**
Each board raises rdy when its partials are ready; the wired-AND all_rdy gives go on both boards in the same cycle.

13 **Chain transfer**
The tail streams its partials, 3 × 32 bits LSB first. The head adds them bit by bit (one full adder + carry) into shreg.

14 **Running argmax**
After each word: D1 = 742 → best_cls 1; D2 = −255, no change; D3 = 765 > 742 → best_cls 3.

15 **Decision**
dec_valid with dec_cls = 3, ≈ 0.75 ms after the last frame. Same class as the golden model for this stride.

Stride 21 (samples 1960–2155): D = head + tail
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D3 = 223 + 542 = 765
argmax{0, D1, D2, D3} = class 3